

Assignment - 1

Course Name: Algebraic Geometry

Indian Statistical Institute, Kolkata

- (1) Let D be an integral domain and let $f(x_1, \dots, x_n) \in D[x_1, \dots, x_n]$. Suppose that there exists an infinite subset $S \subset D$ such that

$$f(a_1, \dots, a_n) = 0 \quad \text{for all } (a_1, \dots, a_n) \in S^n,$$

where $S^n = S \times \dots \times S$ is the n -times Cartesian product of S with itself. Show that f is the zero polynomial.

- (2) Show that the Zariski topology on $\mathbb{A}_k^1$ is Hausdorff if k is a finite field. Show using the definition of Hausdorffness, if k is infinite, the Zariski topology on $\mathbb{A}_k^1$ is not Hausdorff.

- (3) Let $X \subset \mathbb{A}_k^3$ be the union of the three coordinate axes. Determine generators for the ideal $\mathcal{I}(X)$. Show that $\mathcal{I}(X)$ cannot be generated by fewer than 3 elements, although X has codimension 2 in $\mathbb{A}_k^3$ (i.e., $\dim(X) = 1$).

- (4) Let $X = \{(t, t^2, t^3) \mid t \in k\} \subset \mathbb{A}_k^3$. Show that X is an affine variety of dimension 1 and compute $\mathcal{I}(X)$.

- (5) Let $X = \{(t^3, t^4, t^5) \in \mathbb{A}_k^3 \mid t \in k\}$. Prove that X is Zariski closed in $\mathbb{A}_k^3$ and give three polynomials $f, g, h \in k[x, y, z]$ that generate $\mathcal{I}(X)$. Determine the irreducible components of $\mathcal{Z}(f, g)$.

- (6) Assume k is an algebraically closed field. Determine the irreducible components of $\mathcal{Z}(y^4 - x^6, y^3 - xy^2 - yx^3 + x^4)$ in $\mathbb{A}_k^2$.

- (7) (i) If $f \in k[x_1, \dots, x_n]$, explain how the decomposition of $\mathcal{Z}(f)$ into irreducible components is related to the factorization of f into irreducibles.
(ii) If $f \in k[x, y]$ is an irreducible polynomial, prove that the Zariski topology on $\mathcal{Z}(f)$ is the co-finite topology.
(iii) Give an example of closed subsets $Y, Z \subset \mathbb{A}_k^n$, for some n , that are both reducible but such that $Y \cap Z$ is irreducible.

- (8) In affine 4-dimensional space $\mathbb{A}_k^4$ with coordinates x, y, z, t , let X be the union of the two planes

$$X' = \{x = y = 0\} \quad \text{and} \quad X'' = \{z = x - t = 0\}.$$

Compute the ideal $\mathcal{I}(X) \subset k[x, y, z, t]$. For any $a \in k$ let $\mathcal{I}_a \subset \mathbb{A}_k^3$ be the ideal obtained by substituting $t = a$ in $\mathcal{I}(X)$, and let $X_a = \mathcal{Z}(\mathcal{I}_a) \subset \mathbb{A}_k^3$.

Show that the family of algebraic sets X_a with $a \in k$ describes two skew lines in $\mathbb{A}_k^3$ approaching each other until they finally intersect transversely for $a = 0$.

Moreover, show that the ideals $\mathcal{I}_a$ are radical for $a \neq 0$, but not for $a = 0$. Find the elements of $\text{rad}(\mathcal{I}_0) \setminus \mathcal{I}_0$ and interpret them geometrically.

- (9) Let I be an ideal in $k[X_1, \dots, X_n]$. Prove that $\mathcal{Z}(I)$ is a finite set if and only if $k[X_1, \dots, X_n]/I$ is a finite-dimensional vector space over k . Show that if this occurs, the number of points in $\mathcal{Z}(I)$ is at most

$$\dim_k(k[X_1, \dots, X_n]/I).$$

- (10) Let V, W be algebraic sets in $\mathbb{A}_k^n$, with $V \subset W$. Show that each irreducible component of V is contained in some irreducible component of W .

- (11) Find an example of a non-Noetherian ring R such that its prime spectrum $\text{Spec}(R)$ is a singleton set, i.e., in particular, $\text{Spec}(R)$ endowed with the Zariski topology, is a Noetherian topological space.

- (12) Find an example of an irreducible polynomial $f(X, Y) \in \mathbb{R}[X, Y]$ such that $\mathcal{Z}(f)$ is non-empty but reducible. In this case, the ideal generated by f in $\mathbb{R}[X, Y]$ is prime, but $\mathcal{Z}(f)$ is reducible—explain what goes wrong here.

- (13) Let $X \subset \mathbb{A}_k^4$ be the algebraic set given by the equations

$$x^2 - xz = 0, \quad yx - x^2 = 0, \quad z^2 - xz = 0.$$

Find the irreducible components of X . What are their prime ideals?

- (14) Let Y be a subspace of a topological space X . Show that Y is irreducible if and only if the closure of Y in X is irreducible.

- (15) For an algebraically closed field k , describe all Zariski closed irreducible subvarieties of $\mathbb{A}_k^2$ and deduce that $\dim(\mathbb{A}_k^2) = 2$.

- (16) For any ideal $\mathfrak{a}$ in $k[X_1, \dots, X_n]$, prove that a polynomial $h \in \text{rad}(\mathfrak{a})$ if and only if

$$1 \in \langle \mathfrak{a}, 1 - Yh \rangle R,$$

where $R := k[X_1, \dots, X_n, Y]$ and $\langle \mathfrak{a}, 1 - Yh \rangle R$ denotes the ideal of R generated by the elements of $\mathfrak{a}$ and $1 - Yh$.

- (17) (i) Let $(f_1, \dots, f_m)$ be an m -tuple of polynomials in $k[x_1, \dots, x_n]$ and let

$$f : \mathbb{A}_k^n \rightarrow \mathbb{A}_k^m$$

be the map given by

$$P \mapsto (f_1(P), \dots, f_m(P)).$$

Show that f is continuous, taking the Zariski topologies on $\mathbb{A}_k^m$ and $\mathbb{A}_k^n$. Also show, by means of a concrete example, that f need not be a closed map.

- (ii) Let k be an algebraically closed field and $Y \subset \mathbb{A}_k^n$ be an affine variety having co-ordinate ring $A(Y)$. Let $F(Y, k)$ be the k -algebra of k -valued functions on Y . Prove that the natural map

$$k[x_1, \dots, x_n] \rightarrow F(Y, k)$$

given by $f \mapsto f|_Y$ induces an injective homomorphism

$$A(Y) \hookrightarrow F(Y, k).$$

- (iii) Let k be algebraically closed. If $f, g \in F(Y, k)$ are continuous functions (where we give $k = \mathbb{A}_k^1$ the Zariski topology) and $fg = 0$, show that either $f = 0$ or $g = 0$.